CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

31. Match **List I** with **List II**.

	List I		List II
A.	Elastomeric polymer	I.	Urea formaldehyde resin
B.	Fibre polymer	II.	Polystyrene
C.	Thermosetting polymer	III.	Polyester
D.	Thermoplastic polymer	IV.	Neoprene

Choose the correct answer from the options given below?

(1) A-IV, B-III, C-I, D-II

(2) A-II, B-III, C-I, D-IV

(3) A-IV, B-I, C-III, D-II

(4) A-II, B-I, C-IV, D-III

Answer (1)

Sol. A. Elastomeric fibre (IV) Neoprene

B. Fibre polymer (III) Polyester

C. Thermosetting polymer (I) Urea Formaldehyde Resin

D. Thermoplastic polymer (II) Polystyrene

32. A solution of CrO_5 in amyl alcohol has a _____ colour.

(1) Blue (2) Yellow

(3) Green (4) Orange-Red

Answer (1)

Sol. CrO_5 is blue in colour in amyl alcohol.

33. The concentration of dissolved oxygen in water for growth of fish should be more than X ppm and Biochemical Oxygen Demand in clean water should be less than Y ppm. X and Y in ppm are, respectively.

(1) X Y (2) X Y

4 15 6 12

(3) X Y (4) X Y

4 8 6 5

Answer (4)

Sol. For high growth, dissolved oxygen should be less than 6 ppm and clean water has dissolved oxygen less than 5 ppm.

34. Correct order of spin only magnetic moment of the following complex ions is

(Given At.no. Fe:26, Co:27)

(1) $[Co(C_2O_4)_3]^{3-} > [CoF_6]^{3-} > [FeF_6]^{3-}$

(2) $[FeF_6]^{3-} > [Co(C_2O_4)_3]^{3-} > [CoF_6]^{3-}$

(3) $[FeF_6]^{3-} > [CoF_6]^{3-} > [Co(C_2O_4)_3]^{3-}$

(4) $[CoF_6]^{3-} > [FeF_6]^{3-} > [Co(C_2O_4)_3]^{3-}$

Answer (3)

Sol. $[FeF_6]^{3-} \longrightarrow$ 5 unpaired electrons

$[Co(C_2O_4)_3]^{3-} \longrightarrow$ 0 unpaired electron

$[CoF_6]^{3-} \longrightarrow$ 4 unpaired electrons

So, correct answer is : (3)

35. According to MO theory the bond orders for O_2^{2-}, CO and NO^+ respectively, are

(1) 1, 2 and 3 (2) 1, 3 and 2

(3) 2, 3 and 3 (4) 1, 3 and 3

Answer (4)

Sol.

Species	B.O.
O_2^{-2}	1
CO	3
$NO^{\oplus}$	3

36. Match **List-I** with **List-II.**

	List-I		**List-II**
A.	van't Hoff factor, i	I.	Cryoscopic constant
B.	k_f	II.	Isotonic solutions
C.	Solutions with same osmotic pressure	III.	$\frac{\text{Normal molar mass}}{\text{Abnormal molar mass}}$
D.	Azeotropes	IV.	Solutions with same composition of vapour above it

Choose the correct answer from the options given below.

(1) A-III, B-I, C-II, D-IV

(2) A-III, B-I, C-IV, D-II

(3) A-III, B-II, C-I, D-IV

(4) A-I, B-III, C-II, D-IV

Answer (1)

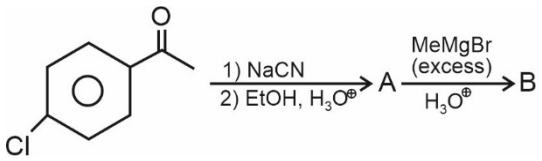
Sol. A. van't Hoff factor — III. $\frac{\text{Normal molar mass}}{\text{Abnormal molar mass}}$

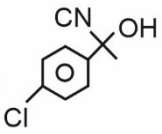
B. k_f — I. Cryoscopic constant

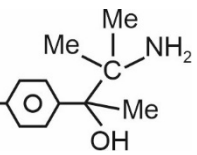
C. Solutions with same osmotic pressure — II. Isotonic solutions

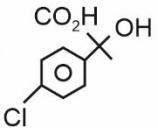
D. Azeotropes — IV. Solutions with same composition of vapour above it

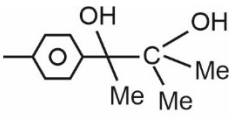
37. Find out the major products from the following reaction sequence.

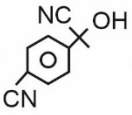


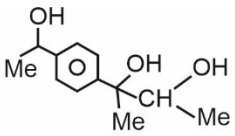
(1) A = 

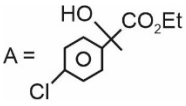
, B = 

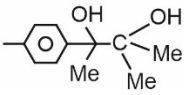
(2) A = 

, B = 

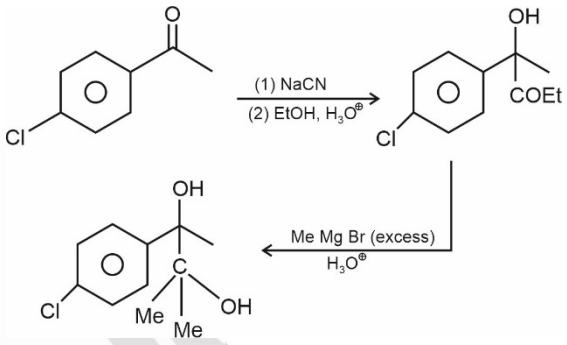
(3) A = 

, B = 

(4) A = 

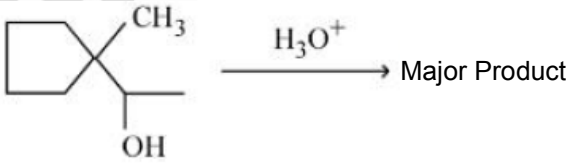
, B = 

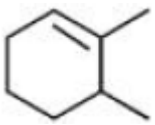
Answer (4)

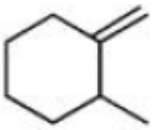
Sol. 

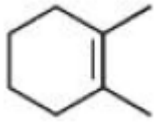
Correct answer is (4)

38. Find out the major product for the following reaction.



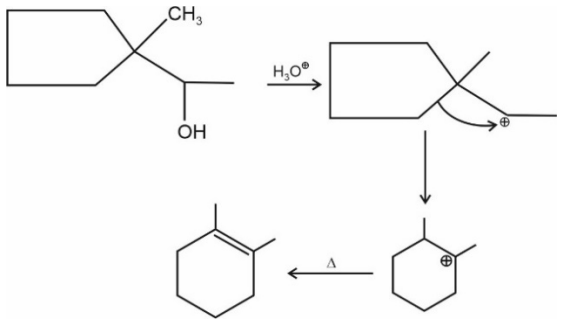
(1) 

(2) 

(3) 

(4) 

Answer (3)

Sol. 

39. The one giving maximum number of isomeric alkenes on dehydrohalogenation reaction is (excluding rearrangement)

(1) 1-Bromo-2-methylbutane

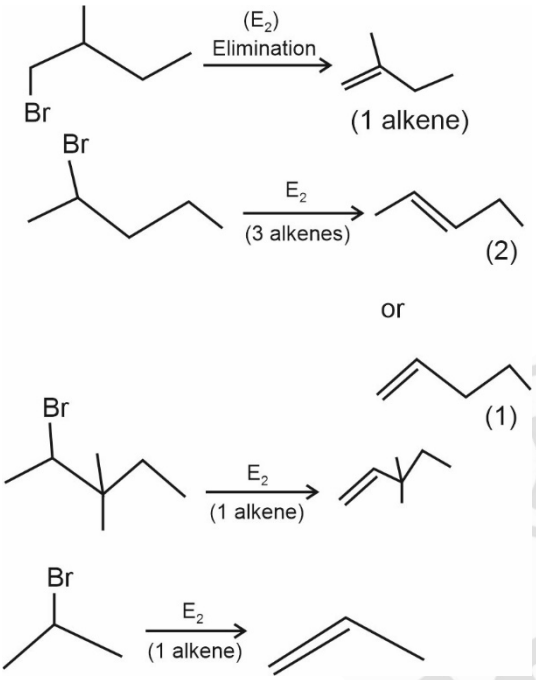
(2) 2-Bromopentane

(3) 2-Bromo-3, 3-dimethylpentane

(4) 2-Bromopropane

Answer (2)

Sol.



40. Given below are two statements:

Statement I : The decrease in first ionization enthalpy from B to Al is much larger than that from Al to Ga.

Statement-II : The d orbitals Ga are in completely filled.

In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both the statements I and II are correct

(4) Both the statements I and II are incorrect

Answer (3)

Sol. Statement I : It is correct as decrease in first ionisation enthalpy is much larger from B to Al.

Statement II : It is also correct

41. An indicator 'X' is used for studying the effect of variation in concentration of iodide on the rate of reaction of iodide ion with H_2O_2 at room temp. The indicator 'X' forms blue colored complex with compound 'A' present in the solution. The indicator 'X' and compound 'A' respectively are

(1) Starch and H_2O_2

(2) Starch and iodine

(3) Methyl orange and H_2O_2

(4) Methyl orange and iodine

Answer (2)

Sol. Starch forms blue coloured complex with iodine.

42. Given below are two statements:

Statement I : Nickel is being used as the catalyst for producing syn gas and edible fats.

Statement II : Silicon forms both electron rich and electron deficient hydrides.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both the statements I and II are incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both the statements I and II are correct

(4) Statement I is correct but statement II is incorrect

Answer (4)

Sol. Statement I is correct.

Statement II is incorrect as Si forms electron precise hydride.

43. A doctor prescribed the drug Equanil to a patient. The patient was likely to have symptoms of which disease?

(1) Hyperacidity

(2) Depression and hypertension

(3) Stomach ulcers

(4) Anxiety and stress

Answer (2)

Sol. Equanil is for depression and hypertension.

44. When a hydrocarbon A undergoes combustion in the presence of air, it requires 9.5 equivalents of oxygen and produces 3 equivalents of water. What is the molecular formula of A?

(1) C_9H_9 (2) C_8H_6

(3) C_6H_6 (4) C_9H_6

Answer (2)

Sol. $C_xH_y + \left(x + \frac{y}{4}\right)O_2 \rightarrow xCO_2 + \frac{y}{2}H_2O$

$\frac{y}{2} = 3$

$y = 6$

$x + \frac{y}{4} = \frac{19}{2}$

$x = \frac{19}{2} - \frac{3}{2} = 8$

So, formula is C_8H_6.

45. Match **List I** and **List II**.

	List I		**List II**
A.	Osmosis	I.	Solvent molecules pass through semi permeable membrane towards solvent side.
B.	Reverse osmosis	II.	Movement of charged colloidal particles under the influence of applied electric potential towards oppositely charged electrodes.
C.	Electro osmosis	III.	Solvent molecules pass through semi permeable membrane towards solution side.
D.	Electrophoresis	IV.	Dispersion medium moves in an electric field.

Choose the correct answer from the options given below :

(1) A-III, B-I, C-IV, D-II (2) A-I, B-III, C-IV, D-II

(3) A-III, B-I, C-II, D-IV (4) A-I, B-III, C-II, D-IV

Answer (1)

Sol. Correct match is:

A.	Osmosis	III.	Solvent molecules pass through semi permeable membrane towards solution side.
B.	Reverse osmosis	I.	Solvent molecules pass to solvent side through SPM.
C.	Electro osmosis	IV.	Dispersion medium moves in an electric field.
D.	Electrophoresis	II.	Movement of colloidal particle towards oppositely charged electrodes.

46. Which of the following relations are correct?

(A) $\Delta U = q + p\Delta V$ (B) $\Delta G = \Delta H - T\Delta S$

(C) $\Delta S = \frac{q_{rev}}{T}$ (D) $\Delta H = \Delta U - \Delta nRT$

Choose the most appropriate answer from the options given below:

(1) B and C Only

(2) C and D Only

(3) A and B Only

(4) B and D Only

Answer (1)

Sol. (A) $\Delta U = q - p\Delta V$

(B) $\Delta G = \Delta H - T\Delta S$

(C) $\Delta S = \frac{q_{rev}}{T}$

(D) $\Delta H = \Delta U + (\Delta nRT)$

Hence, (B) and (C) relations are correct.

47. The major component of which of the following ore is sulphide based mineral?

(1) Sphalerite (2) Calamine

(3) Malachite (4) Siderite

Answer (1)

Sol. Sphalerite $\rightarrow$ ZnS

Calamine $\rightarrow ZnCO_3$

Malachite $\rightarrow CuCO_3 \cdot Cu(H)_2$

Siderite $\rightarrow FeCO_3$

48. Reaction of propanamide with Br_2/KOH(aq) produces :

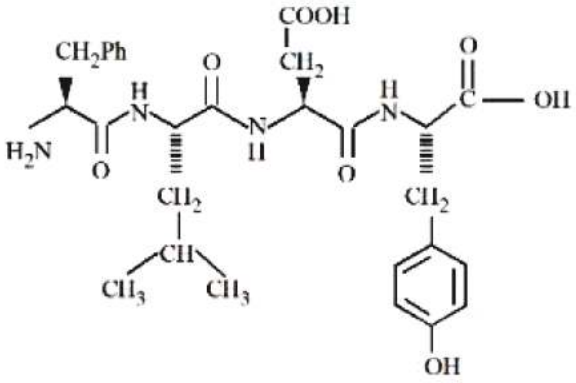
(1) Propanenitrile (2) Ethylnitrile

(3) Ethylamine (4) Propylamine

Answer (3)

Sol. $\text{Propanamide} \xrightarrow{Br_2/KOH} \text{Ethylamine}$

49. Following tetrapeptide can be represented as



(F, L, D, Y, I, Q, P are one letter codes for amino acids)

(1) YQLF (2) FLDY

(3) PLDY (4) FIQY

Answer (2)

Sol. The tetrapeptide code is

FLDY

F $\rightarrow$ Phenyl alanine

L $\rightarrow$ Leucine

D $\rightarrow$ Aspartic acid

Y $\rightarrow$ Tyrosine

50. The set of correct statements is:

(i) Manganese exhibits +7 oxidation state in its oxide.

(ii) Ruthenium and Osmium exhibit +8 oxidation in their oxides.

(iii) Sc shows +4 oxidation state which is oxidizing in nature.

(iv) Cr shows oxidising nature in +6 oxidation state.

(1) (ii) and (iii) (2) (i), (ii) and (iv)

(3) (ii), (iii) and (iv) (4) (i) and (iii)

Answer (2)

Sol. (i) $Mn_2O_7 \rightarrow$ Mn in (+7) oxidation state

(ii) It is also correct

(iii) Sc only shows +3 oxidation state

(iv) Cr^{+6} is oxidising in nature

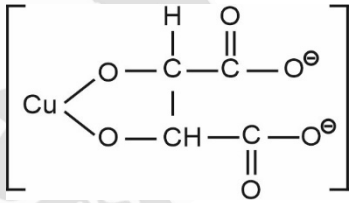
SECTION - B

Numerical Value Type Questions: This section contains 10 questions. In Section B, attempt any five questions out of 10. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the second decimal place; e.g. 06.25, 07.00, –00.33, –00.30, 30.27, –27.30) using the mouse andw the on-screen virtual numeric keypad in the place designated to enter the answer.

51. The denticity of the ligand present in the Fehling's reagent is _____.

Answer (4)

Sol. The denticity is 2 in Fehling' reagent.



However, the overall maximum denticity can be 4 with other metal ion.

52. On heating, $LiNO_3$ gives how many compounds among the following? _____

Li_2O, N_2, O_2, $LiNO_2$, NO_2

Answer (3)

Sol. $LiNO_3 \xrightarrow{\Delta} Li_2O + NO_2 + O_2$

53. The equilibrium constant for the reaction $Zn(s) + Sn^{2+}(aq) \rightleftharpoons Zn^{2+}(aq) + Sn(s)$ is 1×10^{20} at 298 K. The magnitude of standard electrode potential of Sn/Sn^{2+} if $E^{o}_{Zn^{2+}/Zn} = -0.76$ V is _____ $\times 10^{-2}$ V.

(Nearest integer).

Given : $\frac{2.303\,RT}{F} = 0.059$ V

Answer (17)

Sol. $E^{\circ}_{cell} = \frac{2.303\,RT}{2F}\log k$

$$E^{\circ}_{cell} = \frac{0.059}{2}\log(10^{20})$$

$$E^{\circ}_{Zn^{2+}/Zn} + 0.76 = 0.59$$

$$E^{\circ}_{Zn^{2+}/Zn} = 0.59 - 0.76$$

$$E^{\circ}_{Zn/Zn^{2+}} = 0.17\ V$$

54. A metal M forms hexagonal close-packed structure. The total number of voids in 0.02 mol of it is _____ × 10^{21} (Nearest integer).
(Given $N_A = 6.02 \times 10^{23}$)

Answer (36)

Sol. Number of voids = $0.02 \times 3 \times 6.02 \times 10^{23}$

$\simeq 36 \times 10^{21}$

55. When 0.01 mol of an organic compound containing 60% carbon was burnt completely, 4.4 g of CO_2 was produced. The molar mass of compound is _____ g mol^{-1} (Nearest integer).

Answer (200)

Sol. Number of moles of $CO_2 = \frac{4.4}{44} = 0.1$

$\therefore$ Number of moles of C in 1 mole of compound

= 10

$\therefore$ $120 = \frac{60}{100} \times (x)$ [where x is molar mass of OC]

Molar mass = 200 g mol^{-1}

56. Total number of acidic oxides among N_2O_3, NO_2, N_2O, Cl_2O_7, SO_2, CO, CaO, Na_2O and NO is _____.

Answer (4)

Sol. N_2O_3, NO_2, Cl_2O_7, SO_2 are acidic in nature

57. For conversion of compounds A $\rightarrow$ B, the rate constant of the reaction was found to be 4.6 × 10–5 L mol^{-1} s^{-1}. The order of the reaction is_____.

Answer (2)

Sol. As unit is L mol^{-1} s^{-1}, order of the reaction is 2.

58. Assume that the radius of the first Bohr orbit of hydrogen atom is 0.6 Å. The radius of the third Bohr orbit of He^+ is _____ picometer. (Nearest Integer)

Answer (270)

Sol. Radius of 3rd Bohr orbit of He^{+3}

$= 0.6 \times \frac{(3)^2}{2}$

= 0.3 × 9

= 2.7 Å

= 270 × 10^{-12} pm

59. At 298 K

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH3(g), K_1 = 4 \times 10^5$$

$$N_2(g) + O_2(g) \rightleftharpoons 2NO(g), K_2 = 1.6 \times 10^{12}$$

$$H_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons H_2O(g), K_3 = 1.0 \times 10^{-13}$$

Based on above equilibria, the equilibrium constant of the reaction,

$$2NH_3(g) + \frac{5}{2}O_2(g) \rightleftharpoons 2NO(g) + 3H_2O(g)$$

is_____ × 10^{-33} (Nearest integer).

Answer (4)

Sol. $2NH_3(g)\frac{5}{2}O_2(g) \rightleftharpoons 2NO(g) + 3H_2O(g)$

Clearly, $K_{eq} = \frac{1}{k_1} \times k_2 \times k_3^3$

$= \frac{1.6 \times 10^{12} \times 10^{-39}}{4 \times 10^5}$

= 0.4 × 10^{-32}

= 4 × 10^{-33}

60. The volume of HCl, containing 73 g L^{-1}, required to completely neutralise NaOH obtained by reacting 0.69 g of metallic sodium with water, is _____ mL. (Nearest Integer) (Given: molar Masses of Na, Cl, O, H, are 23, 35.5, 16 and 1 g mol^{-1} respectively)

Answer (15)

Sol. $\underset{0.69\,g}{Na} + H_2O \rightarrow \underset{0.03\,moles}{NaOH} + \frac{1}{2}H_2$

= 0.03 moles

$\therefore$ 0.03 = 2 × V

$V = \frac{0.03}{2}L$

= 15 mL

MATHEMATICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

61. The area of the region

$$A=\left\{(x,\ y):|\cos x-\sin x|\le y\le \sin x,\ 0\le x\le \frac{\pi}{2}\right\}$$

is

(1) $\sqrt{5}-2\sqrt{2}+1$

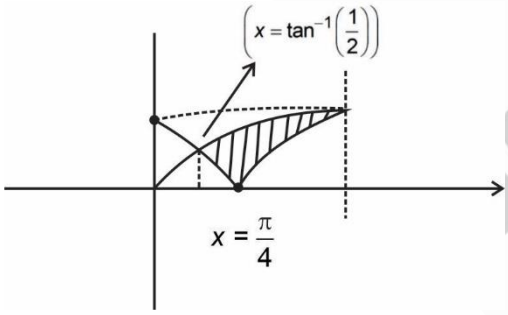
(2) $\frac{3}{\sqrt{5}}-\frac{3}{\sqrt{2}}+1$

(3) $\sqrt{5}+2\sqrt{2}-4.5$

(4) $1-\frac{3}{\sqrt{2}}+\frac{4}{\sqrt{5}}$

Answer (1)

Sol.



$$\text{Area}=\int_{\tan^{-1}\frac{1}{2}}^{\frac{\pi}{4}}\left(\sin x-(\cos x-\sin x)\right)dx+\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\left(\sin x-(\sin x-\cos x)\right)dx$$

$$=\int_{\tan^{-1}\frac{1}{2}}^{\frac{\pi}{4}}(2\sin x-\cos x)\,dx+(\sin x)_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$=2\cos x-\sin x\Big|_{\tan^{-1}\frac{1}{2}}^{\frac{\pi}{4}}+\left(1-\frac{1}{\sqrt{2}}\right)$$

$$=\frac{3}{\sqrt{2}}+\frac{1}{\sqrt{5}}+\frac{1}{\sqrt{5}}+1-\frac{1}{\sqrt{2}}$$

$$=\sqrt{5}-2\sqrt{2}+1$$

62. Let K be the sum of the coefficients of the odd powers of x in the expansion of $(1+x)^{99}$. Let a be the middle term in the expansion of $\left(2+\frac{1}{\sqrt{2}}\right)^{200}$. If $\frac{{}^{200}C_{99}K}{a}=\frac{2^l m}{n}$, where m and n are odd numbers, then the ordered pair (l, n) is equal to

(1) (51, 101) (2) (51, 99)

(3) (50, 101) (4) (50, 51)

Answer (3)

Sol. $K=2^{98}$

$a={}^{200}C_{100}\ 2^{50}$

$$\therefore\quad \frac{{}^{200}C_{99}\cdot 2^{98}}{{}^{200}C_{100}\cdot 2^{50}}=\frac{2^l m}{n}$$

$$\Rightarrow \frac{100}{101}\cdot 2^{48}=\frac{2^l m}{n}$$

$$\Rightarrow \frac{25}{101}\cdot 2^{50}=\frac{2^l m}{n}$$

$\therefore\ l=50,\ m=25,\ n=101$

63. If the lines $\frac{x-1}{1}=\frac{y-2}{2}=\frac{z+3}{1}$ and $\frac{x-a}{2}=\frac{y+2}{3}=\frac{z-3}{1}$ intersect at the point P, then the distance of the point P from the plane $z=a$ is

(1) 28 (2) 16

(3) 10 (4) 22

Answer (1)

Sol. $\frac{x-1}{1}=\frac{y-2}{2}=\frac{z+3}{1}=\lambda$ (say)

& $\frac{x-a}{2}=\frac{y+2}{3}=\frac{z-3}{1}=\mu$ (say)

$\therefore\ \lambda+1=2\mu+a$ ...(i)

$2\lambda+2=3\mu+2$ ...(ii)

$\lambda-3=\mu+3$ ...(iii)

By (i) & (ii)

$\Rightarrow\ 3\mu-2=4\mu+2a+2$